Activity 3 - Quantify a Base-Year Inventory and Reporting Guide

Alignment: K1 | K2 | K5 | A3 | A7 | LO1 Duration: 75 minutes Objective: Apply activity data, emission factors, GWP values, uncertainty judgement and base-year recalculation rules to produce a transparent inventory.

Scenario

EcoForge has electricity, gas, diesel, refrigerant top-up, purchased aluminium, inbound freight and business-travel data. Units are inconsistent, one electricity month is estimated, the refrigerant factor source is unclear and an acquired line will enter the organisation in 2026. Management wants 2025 as the base year and a rule for future structural and methodological changes.

Materials

- Activity-data pack
- Illustrative emission-factor and GWP register
- Calculation and uncertainty workbook
- Base-year policy template

Questions

- What conversions and unit controls are required?
- Which factor vintage and GWP basis will be disclosed?
- How should uncertainty and estimated data affect decisions?
- What events trigger base-year recalculation?

Detailed procedure

1. Validate source documents, reporting period, units and completeness before calculating.
2. Convert all activity data into factor-compatible units and retain the conversion trail.
3. Apply the equation activity data multiplied by emission factor multiplied by any required GWP conversion.
4. Keep biogenic carbon dioxide, removals and purchased offsets separately identified from inventory emissions.
5. Reconcile totals to source documents and investigate the missing electricity month and refrigerant-factor provenance.
6. Rate uncertainty qualitatively for activity data, factors, models and boundary decisions; propose proportionate improvements.
7. Draft a base-year recalculation policy for acquisitions, divestments, methodology changes and significant error corrections.
8. Draft reporting rules covering boundaries, methods, factor sources, significant exclusions, uncertainty and changes.
9. Perform an independent peer recalculation of two rows and resolve differences.

Required evidence

- Quantified inventory workbook

- Factor and GWP register
- Uncertainty improvement log
- Base-year and reporting guideline

Acceptance check

Every result traces to source data, units, factor and GWP basis; totals reconcile; uncertainty and recalculation triggers are explicit; no offset is netted into the inventory.

Files

- Activity-03-Workbook.xlsx - editable working file
- Activity-03-Activity-Pack.pdf - printable scenario, questions, procedure and checklist

Use fictional, anonymised or authorised workplace information only.